

**Multiple Choice:**

1. The volume of water in a spherical tank having a diameter of 4 m is 5.236 m^3 . Determine the depth of the water in the tank.
A. 1.0
B. 1.2
C. 1.4
D. 1.8
2. Each side of a cube is increased by 1%. By what percent is the volume of the cube increased?
A. 1.21 %
B. 2.8 %
C. 3.03 %
D. 3.5 %
3. Find the volume of a cone to be constructed from a sector having a diameter of 72 cm and a central angle of 150° .
A. $15,533.32 \text{ cm}^3$
B. $16,622.44 \text{ cm}^3$
C. $13,503.44 \text{ cm}^3$
D. $18,866.44 \text{ cm}^3$
4. The angle subtended by an arc is 24° . If the radius of the circle is 45 cm, find the length of arc.
A. 16.85 cm
B. 17.85 cm
C. 18.85 cm
D. 19.85 cm
5. A certain angle has a supplement 5 times its complement. What is the angle?
A. 67.5°
B. 58.5°
C. 30°
D. 27°
6. How many diagonals are there in a polygon of 20 sides?
A. 200
B. 170
C. 100
D. 158
7. A circle with radius 6 cm has half its area removed by cutting off a border of uniform width. Find the width of the border.
A. 1.76 cm
B. 1.35 cm
C. 1.98 cm
D. 2.03 cm
8. The area of a triangle inscribed in a circle is 39.19 cm^2 and the radius of the circumscribed circle is 7.14 cm. If the two sides of the inscribed triangle are 8 cm and 10 cm, respectively, find the third side.
A. 11 cm
B. 12 cm
C. 13 cm
D. 14 cm
9. The distance between the centers of the three circles which are mutually tangent to each other externally are 10, 12 and 14 units. The area of the largest circle is
A. 72π
B. 23π
C. 64π
D. 16π
10. A rhombus has diagonals of 32 and 20 inches. Determine its area.
A. 360 in^2
B. 280 in^2
C. 320 in^2
D. 400 in^2
11. Find the area of a regular hexagon inscribed in a circle of radius 1.
A. 1.316
B. 2.945
C. 2.598
D. 3.816
12. Determine the area of a regular 6-star polygon if the inner regular hexagon has 10 cm sides.
A. 441.66 cm^2
B. 467.64 cm^2
C. 519.60 cm^2
D. 493.62 cm^2
13. Find the area of a regular pentagon whose side is 25 m and apothem is 17.2 m.
A. 1075 m^2
B. 1085 m^2
C. 1080 m^2
D. 1095 m^2
14. How many sides are in a polygon if each interior angle is 165 degrees?
A. 12
B. 24
C. 20
D. 48
15. Given a sphere of diameter, d. What is the percentage increase in its volume when the surface area increases by 21 %?
A. 5 %
B. 10 %
C. 21 %
D. 33 %
16. A regular triangular pyramid has an altitude of 9 m and a volume of 187.06 cu. m . What is the base edge in meters?
A. 12
B. 13
C. 14
D. 15
17. The central angle of a spherical wedge is 1 radian. Find its volume if its radius is 1 unit.
A. $2/3$
B. $1/2$
C. $3/4$
D. $2/5$
18. The bases of a right prism are hexagons with one of each side equal to 6 cm. The bases are 12 cm apart. What is the volume of the right prism?
A. 1211.6 cm^3
B. 2211.7 cm^3
C. 1212.5 cm^3
D. 1122.4 cm^3



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20. A conical vessel has a height of 24 cm and a base diameter of 12 cm. It holds water to a depth of 18 cm above its vertex. Find the volume (in cm^3) of its content.
- A. 188.40
B. 298.40
C. 381.70
D. 412.60
21. A circular piece of cardboard with a diameter of 1 m will be made into a conical hat 40 cm high by cutting a sector off and joining the edges to form a cone. Determine the angle subtended by the sector removed.
- A. 144°
B. 148°
C. 152°
D. 154°
22. Find the angle in mils subtended by a line 10 yards long at a distance of 5000 yards.
- A. 1
B. 2
C. 3
D. 4
23. A rat fell on a bucket of a water wheel with diameter of 600 cm which traveled an angle of 190° before it dropped from the bucket. Calculate for the linear cm that the rat was carried by the bucket before it fell.
- A. 950
B. 965
C. 985
D. 995
24. If the sides of a parallelogram and an included angle are 6, 10 and 100° , respectively, find the length of the shorter diagonal.
- A. 10.63
B. 10.37
C. 10.73
D. 10.23
25. Find the difference of the area of the square inscribed in a semi-circle having a radius of 15 m. The base of the square lies on the diameter of the semi-circle.
- A. 171.5 cm^2
B. 172.5 cm^2
C. 173.5 cm^2
D. 174.5 cm^2
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